Chapter 10. Analysis of Variance and Regression

One-way and two-way ANOVA, quadratic forms, linear regression

Contents

10.1 One-way analysis of variance **2**

10.2 Two-way analysis of variance **3**

10.3 Quadratic forms of normal vectors **3**

10.4 Linear regression **3**

10.1 One-way analysis of variance

Definition (One-way normal model)

$X_{ij} = \mu_i + e_{ij}$ with $e_{ij} \overset{\text{i.i.d.}}{\sim} N(0,\sigma^2)$, $i = 1,\dots,k$, $j = 1,\dots,n_i$, $n = \sum n_i$. Writing $\bar\mu = \sum n_i\mu_i/n$ and $\alpha_i = \mu_i - \bar\mu$ (so $\sum n_i\alpha_i = 0$), the model becomes $X_{ij} = \bar\mu + \alpha_i + e_{ij}$.

Theorem 10.1

(a) $(\bar X_1,\dots,\bar X_k, \sum_{i,j}(X_{ij}-\bar X_i)^2)$ is complete sufficient, so $\hat\mu_i = \bar X_i$, $\hat\alpha_i = \bar X_i - \bar X$ and $\hat\sigma^2 = SSE/(n-k)$ are UMVUEs, where $SSE = \sum_{i,j}(X_{ij}-\bar X_i)^2$. (b) The $\bar X_i \sim N(\mu_i,\sigma^2/n_i)$ are independent and independent of SSE, and $SSE/\sigma^2 \sim \chi^2(n-k)$.

Proof. (a) The joint density is a $(k+1)$-parameter exponential family in these statistics; apply Lehmann–Scheffé (Theorem 8.3). (b) Apply Theorem 4.2 of Volume I to each group and use independence across groups; chi-squares add. ∎

Decomposition and the F-test. With $SST = \sum_{i,j}(X_{ij}-\bar X)^2$ and $SSTr = \sum_i n_i(\bar X_i - \bar X)^2$,

$$SST = SSTr + SSE,$$

since the cross term $\sum_i (\bar X_i - \bar X)\sum_j (X_{ij}-\bar X_i)$ vanishes.

Theorem 10.2

The likelihood ratio test of $H_0: \alpha_1 = \cdots = \alpha_k = 0$ rejects when

$$F = \frac{SSTr/(k-1)}{SSE/(n-k)} \ge F_\alpha(k-1,n-k),$$

and under H_0, $F \sim F(k-1,n-k)$.

Proof. The restricted and full MLEs of σ^2 are SST/n and SSE/n, so $2\log\Lambda = n\log(SST/SSE) = n\log\left(1 + \frac{k-1}{n-k}F\right)$, which is increasing in F. Under H_0, $SSTr/\sigma^2 = \|(P_G - P_1)\mathbf{e}\|^2/\sigma^2$, where P_G projects onto group-mean vectors and P_1 onto constants. $P_G - P_1$ is idempotent of rank $k-1$ and orthogonal to $I - P_G$, so Theorem 4.9 of Volume I gives independent $\chi^2(k-1)$ and $\chi^2(n-k)$. ∎

Simultaneous intervals. For contrasts $\sum c_i\mu_i$ with $\sum c_i = 0$:

- **Scheffé**: $\sum c_i\bar X_i \pm \sqrt{(k-1)F_\alpha(k-1,n-k)}\,\hat\sigma\sqrt{\sum c_i^2/n_i}$ holds simultaneously for *all* contrasts with confidence $1-\alpha$.
- **Bonferroni**: for a fixed list of m comparisons, use $t_{\alpha/(2m)}(n-k)$ in each; this is shorter when m is small.

Why Scheffé works. By Cauchy–Schwarz, $\left(\sum_i c_i(\bar X_i - \mu_i)\right)^2 \le \sum_i \frac{c_i^2}{n_i}\cdot\sum_i n_i\{(\bar X_i - \bar X) - (\mu_i - \bar\mu)\}^2$ for contrasts (subtracting the common $\bar X - \bar\mu$ does not change a contrast). The last sum is $\sigma^2\chi^2(k-1)$, independent of $\hat\sigma^2$, so the bound holds for all $\mathbf{c}$ simultaneously with probability $P(F(k-1,n-k) \le F_\alpha) = 1-\alpha$. ∎

10.2 Two-way analysis of variance

With r replications per cell, $X_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}$, $i = 1, \dots, a$, $j = 1, \dots, b$, $k = 1, \dots, r$, with the side conditions $\sum_i \alpha_i = \sum_j \beta_j = \sum_i \gamma_{ij} = \sum_j \gamma_{ij} = 0$. The orthogonal decomposition

$$SST = SSA + SSB + SSAB + SSE$$

uses $SSA = br \sum_i (\bar{X}_{i\cdot\cdot} - \bar{X})^2$, $SSB = ar \sum_j (\bar{X}_{\cdot j\cdot} - \bar{X})^2$, $SSAB = r \sum_{i,j} (\bar{X}_{ij\cdot} - \bar{X}_{i\cdot\cdot} - \bar{X}_{\cdot j\cdot} + \bar{X})^2$ and $SSE = \sum (X_{ijk} - \bar{X}_{ij\cdot})^2$. The degrees of freedom are $a-1$, $b-1$, $(a-1)(b-1)$ and $ab(r-1)$. Each null hypothesis (no A effect, no B effect, no interaction) is tested by the corresponding mean square over $SSE/\{ab(r-1)\}$, which has an F distribution under that null. The proof is the same projection argument: the four pieces are squared lengths of projections onto mutually orthogonal subspaces. With $r = 1$ there is no SSE; assume $\gamma_{ij} = 0$ and use $SSAB$ as the error.

10.3 Quadratic forms of normal vectors

Theorem 10.3 (Cochran)

Let $\mathbf{Z} \sim N_n(\mathbf{0}, I)$ and $I = A_1 + \cdots + A_m$ with symmetric A_j of ranks r_j. If $\sum r_j = n$, then the $\mathbf{Z}' A_j \mathbf{Z}$ are independent with $\chi^2(r_j)$ distributions.

Proof. $\sum r_j = n$ forces each A_j to be idempotent and $A_j A_l = 0$ for $j \neq l$. (Write each A_j as a sum of r_j rank-one terms $\pm \mathbf{v}\mathbf{v}'$; the n vectors together form a matrix V with $V \,\mathrm{diag}(\pm 1) V' = I$, which forces all signs $+$ and V orthogonal.) Then Theorem 4.9 of Volume I applies. ∎

If $\mathbf{Y} \sim N_n(\mu, \sigma^2 I)$ and A is a projection of rank r, then $\mathbf{Y}' A \mathbf{Y}/\sigma^2$ has a **noncentral** $\chi^2(r, \lambda)$ distribution with $\lambda = \mu' A \mu / \sigma^2$. Its mean is $r + \lambda$, which is why F tests have power increasing in λ.

10.4 Linear regression

In $\mathbf{Y} = X\beta + \mathbf{e}$ with $\mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I)$ and $\operatorname{rank} X = p + 1$:

Theorem 10.4

(a) $\hat{\beta} = (X'X)^{-1} X'\mathbf{Y} \sim N(\beta, \sigma^2 (X'X)^{-1})$; (b) $SSE = \|\mathbf{Y} - X\hat{\beta}\|^2$ satisfies $SSE/\sigma^2 \sim \chi^2(n - p - 1)$ and is independent of $\hat{\beta}$; (c) $\hat{\beta}$ and $\hat{\sigma}^2 = SSE/(n-p-1)$ are UMVUEs (Gauss–Markov: even without normality, $\mathbf{c}'\hat{\beta}$ has the smallest variance among linear unbiased estimators of $\mathbf{c}'\beta$).

Proof. (a) $\hat{\beta}$ is linear in a normal vector. (b) $SSE = \mathbf{e}'(I - \Pi)\mathbf{e}$, a projection of rank $n - p - 1$. $\hat{\beta} - \beta = (X'X)^{-1}X'\mathbf{e}$ and $(X'X)^{-1}X'(I - \Pi) = 0$, so they are independent. (c) Completeness of $(X'\mathbf{Y}, \mathbf{Y}'\mathbf{Y})$ in the exponential family. For Gauss–Markov, any linear unbiased $\mathbf{a}'\mathbf{Y}$ has $X'\mathbf{a} = \mathbf{c}$; write $\mathbf{a} = X(X'X)^{-1}\mathbf{c} + \mathbf{d}$ with $X'\mathbf{d} = 0$. Then $\operatorname{Var}(\mathbf{a}'\mathbf{Y}) = \sigma^2(\mathbf{c}'(X'X)^{-1}\mathbf{c} + \|\mathbf{d}\|^2)$. ∎

Consequently $(\hat{\beta}_j - \beta_j)/\{\hat{\sigma}\sqrt{(X'X)^{-1}_{jj}}\} \sim t(n-p-1)$. For a submodel with q fewer columns,

$$F = \frac{(SSE_0 - SSE)/q}{SSE/(n-p-1)} \sim F(q, n-p-1)$$

under the null, which is the LRT. At a new point $\mathbf{x}_0$, the interval for the mean response is $\mathbf{x}_0'\hat{\beta} \pm t_{\alpha/2}\hat{\sigma}\sqrt{\mathbf{x}_0'(X'X)^{-1}\mathbf{x}_0}$. The prediction interval for a new observation adds 1 under the root.

The one-way model fits the same framework, with an $n \times (k+1)$ design matrix of rank k made identifiable by the constraint $\sum n_i \alpha_i = 0$.